

Children’s knowledge of *negative* free choice inferences

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Abstract

Sentences of the form $\Diamond(A \vee B)$, such as *Mia is allowed to buy the apple or the banana*, give rise to a free choice inference (FCI), namely the inference that each disjunct is individually permitted, i.e. $\Diamond A \wedge \Diamond B$. Previous studies have shown that children as young as five years of age compute such inferences. In this paper, we investigate *negative* free choice inferences (NFCIs). Sentences of the form $\neg\Box(A \wedge B)$, such as *Mia doesn’t have to buy both the apple and the banana*, give rise to the inference that each of the conjuncts is not required, i.e. $\neg\Box A \wedge \neg\Box B$. Adults reportedly derive such inferences less robustly than positive FCIs, a result that has been taken to support hybrid approaches on which positive FCIs are derived by a non-implicature mechanism, while NFCs are genuine implicatures. Given children’s documented difficulties with scalar implicatures, this approach predicts that they will likewise have difficulty deriving NFCs. We report two truth-value judgment experiments conducted with English-speaking adults and 5–7-year-old children, comparing positive and negative FCIs with positive and negative scalar implicatures. Experiment 1 tested modal statements; Experiment 2 extended the same contrasts to the nominal domain. In both experiments, children were largely adult-like with respect to positive FCIs. However, their behavior with NFCs did not conform to the predictions of the hybrid approach. Although our findings replicate the polarity asymmetry previously observed in adults, we argue that the developmental data ultimately challenge the hybrid account of FCIs.

Keywords: free choice, negative free choice, scalar implicatures, polarity effects, hybrid account, disjunction, nominal quantifiers, language acquisition

1 Introduction

Sentences of the form $\Diamond(A \vee B)$, such as (1), give rise to a so-called *free choice inference* (FCI): the inference that each disjunct is individually permitted. In (1), for instance, the sentence is naturally understood as conveying that Mia is allowed to buy the apple and she is allowed to buy the banana (schematically, $\Diamond A \wedge \Diamond B$).

- (1) Mia is allowed to buy either the apple or the banana.
 $\rightsquigarrow_{FC}$ *Mia is allowed to buy the apple and she is allowed to buy the banana*

Free choice inferences are surprising in light of the classical semantics of modality and disjunction, and they have accordingly given rise to a large theoretical literature. One influential line of analysis treats FCIs as a kind of scalar implicature (Fox 2007; Bar-Lev and Fox 2020; Klinedinst 2007; Chemla 2009; Franke 2013; Santorio and Romoli 2017). On this view, FCIs should, in important respects, pattern with other scalar implicatures like the one illustrated below:

- (2) Mia is allowed to buy the apple.
 $\rightsquigarrow_{SI}$ *Mia doesn't have to buy the apple*

In recent years, however, this implicature-based approach has come under pressure from two directions. The first challenge comes from studies of child language acquisition, many of which show that children as young as five years of age successfully compute FCIs from disjunctive modal statements (Zhou et al. 2013; Tieu et al. 2016; Bill 2017; Huang and Crain 2020; Bleotu et al. 2024; Cochard 2025; Cochard et al. 2025; Bleotu et al. 2026). This is a striking finding, since children are well known to experience difficulties with standard scalar implicatures. If FCIs are a type of scalar implicature, then their early mastery, specifically at ages when they perform relatively less well on scalar implicatures, is unexpected.

The second challenge comes from facts about polarity. In addition to the positive free choice pattern in (1), there are also *negative* free choice inferences (NFCI), which arise with negated universal modal statements embedding conjunction, as in (3). Such sentences give rise to the inference that neither conjunct is individually required (Fox 2007). In (3), the relevant interpretation is that Mia does not have to buy the apple and she does not have to buy the banana (schematically, $\neg\Box A \wedge \neg\Box B$).

- (3) Mia doesn't have to buy both the apple and the banana.
 $\rightsquigarrow_{NFCI}$ *Mia doesn't have to buy the apple and she doesn't have to buy the banana*

If positive and negative FCIs are derived by the same implicature-based mechanism, they might be expected to be similarly robust. Experimental work with adults suggests otherwise. Marty et al. (2021) report that adults do derive negative FCIs, but that these are substantially less robust than positive FCIs. Crucially, they also show that this asymmetry cannot be reduced to a general effect of negation, since 'negative' scalar implicatures, like (4) (sometimes also called *indirect* scalar implicatures), pattern differently.

- (4) Mia doesn't have to buy the apple.
 $\rightsquigarrow_{SI}$ *Mia is allowed to buy the apple*

On the basis of these findings, Marty et al. (2021) and subsequent work propose a *hybrid* account, according to which positive FCIs are derived by a non-implicature mechanism, while negative FCIs are genuine implicatures.

This proposal is especially interesting from an acquisition perspective. The hybrid approach leads to a clear developmental prediction: while there is no expectation that children will behave uniformly with FCIs and regular scalar implicatures (SIs), they should treat NFCIs like SIs, to the extent that these rely on the same mechanisms as ordinary implicatures. In other words, if NFCIs are genuine SIs, children’s behavior on them should align closely with their behavior on SIs.

More specifically, following Marty et al. (2021), in order to control for the potential role of negation, the relevant comparisons are between FCIs and positive SIs such as (2), on the one hand, and NFCIs and negative SIs, such as (4), on the other. These are the comparisons we take up in the present paper.

Experiment 1 tests these predictions of the hybrid approach by comparing children and adults on positive and negative FCIs, alongside positive and negative SIs. Such a design allows us to address both challenges to the implicature approach at once: the developmental asymmetry between FCIs and standard implicatures, and the polarity asymmetry between positive and negative FCIs. Experiment 2 extends the investigation to the nominal domain, where positive and negative FCIs are examined with two goals in mind. First, we ask whether the asymmetry observed by Marty et al. (2021) in adults between positive and negative FCIs also arises in the nominal domain, and indeed whether negative FCIs are attested there at all. Second, taking the adult results as a point of comparison, we test in the nominal domain the same prediction for children. Both experiments further investigate the contrast between positive and negative SIs, which has itself emerged as an interesting issue in the literature, with some studies suggesting that both adults and children may derive negative SIs more robustly than positive ones (Ramotowska et al. 2024; Marty et al. 2024a; Marty et al. 2024b; Marty et al. 2021; Bill et al. 2016).

To preview our findings, the results reproduce and extend the two main challenges to the implicature approach. Children are adult-like with positive FCIs and, just like adults, they derive positive FCIs more robustly than NFCIs. Furthermore, this polarity contrast is found to extend to the nominal domain, where the NFCI is again much weaker than the positive FCI, indeed too weak for us to find clear evidence for it.

At the same time, our findings challenge the hybrid approach. On that view, children should pattern with adults on positive FCIs but diverge more clearly on NFCIs, in line with their difficulties with scalar implicatures. Yet this is not the pattern we observe. In Experiment 1, children compute NFCIs at ceiling, nearly as often as positive FCIs and more often than adults do; moreover, their performance on NFCIs matches their performance on negative SIs. In Experiment 2, however, this pattern disappears: neither adults nor children compute NFCIs, and children’s performance on NFCIs no longer patterns with negative SIs. Taken together, these results do not reveal the clear developmental alignment between NFCIs and negative SIs predicted by the hybrid account. We therefore conclude that, although our findings replicate the polarity asymmetry motivating the hybrid account, the developmental pattern observed in children challenges it.

2 Background

2.1 Free choice inferences as implicatures

As mentioned above, a disjunction embedded under an existential modal, as in (1), gives rise to a conjunctive inference about the two disjuncts. This is unexpected in light of the classical semantics of modals and disjunction, and it has prompted a wide range of theoretical accounts since FCIs were first discussed in the 1970s (von Wright 1970; Kamp 1973).¹

One influential approach to the free choice puzzle treats FCIs as a kind of scalar implicature Fox (2007). Without going into too much detail, the core idea of the implicature approach, in its original formulation by Fox (2007), is that FCIs arise through a process of implicature reasoning that compares (1) with relevant alternatives, such as those in (5). Adding the negation of these alternatives to the asserted sentence yields the inference that Mia is allowed to buy the apple and that she is allowed to buy the pear.

- (5) a. Mia is only allowed to buy the apple.
b. Mia is only allowed to buy the pear.

The implicature approach successfully captures the basic free choice pattern as well as a range of more complex data. More recently, however, it has been challenged by two main lines of work: one arising from acquisition studies, and the other from experimental work with adults. We turn to each in turn, reviewing the relevant previous studies to provide context for the current study.

2.2 Two challenges

2.2.1 The acquisition of FCIs

The first challenge comes from acquisition and is based on a straightforward prediction of the implicature account: if FCIs are scalar implicatures, they should pattern with other examples of SIs. In particular, they should behave like standard cases such as (2).

From an acquisition perspective, this prediction translates into the expectation that children should acquire FCIs along a similar developmental trajectory as regular SIs such as (2). As noted above, this prediction has been investigated in a growing body of work (Zhou et al., 2013; Tieu et al., 2016; Bill, 2017; Huang et al., 2018; Huang and Crain, 2020; Huang et al., 2020; An et al., 2020; Bleotu et al., 2024; Cochar, 2025). Many of these studies report that children around the age of 4–5 years perform at or near adult levels with FCIs, but not with standard SIs.

Zhou et al. (2013) and Tieu et al. (2016) set out to investigate the scalar implicature account of free choice inferences, studying 4-year-old Mandarin-speaking children’s ability to compute FCIs from disjunctive modal statements containing the disjunction *huozhe* and the modal verb

¹Part of the challenge is to explain not only the positive case, but also the fact that the negation of (1), illustrated in (i), is not interpreted simply as the negation of the with-FC reading. Rather, it receives a stronger interpretation, namely that Mia is not allowed to buy one and not allowed to buy the other, a reading sometimes referred to as *double prohibition*. Given that double prohibition is not the focus of the present study, we set it aside here.

- (i) Mia is not allowed to buy either the apple or the pear.
↪ *Mia is not allowed to buy the apple and she is not allowed to buy the pear*

keyi ‘may’, compared to scalar implicature controls containing existential modals (triggering a ‘not required to’ inference) and plain disjunction (triggering an exclusive ‘not both’ inference). Participants (mean age 4;03) completed a Truth Value Judgment Task in which they had to judge a puppet’s description of events. On an example critical FC trial, for example, Kung Fu Panda and Batman were participating in a racing competition, and the judge issued particular rules: Kung Fu Panda was only allowed to push the green car, while Batman was only allowed to push the orange car. A puppet would then provide a reminder of the rules, producing the disjunctive target in (6).

- (6) Gongfu xiongmao keyi tui lüse xiaochē huozhe juse xiaochē
 KungFu Panda may push green car or orange car
 ‘Kung Fu Panda may push the green car or the orange car.’

If the participant computed the FCI, they were expected to reject the sentence, and indeed children rejected the free choice targets 91% of the time, in contrast to only 18% of the time on the SI controls. Tieu et al. took the stark contrast in performance in the two conditions to present a challenge to the implicature account, though they ultimately argued that the implicature account could be preserved on the supposition that the alternatives required to compute FCIs, being present in the test sentences themselves, could facilitate children’s performance, in contrast to cases where the required scalar alternatives are lexical in nature.

In a partial replication of Tieu et al.’s study, Experiment 1 of Huang and Crain (2020) also investigated Mandarin-speaking children’s interpretation of disjunctive modal statements in a Truth Value Judgment Task. In one target story, for example, a small dog and a big dog were athletes and Mr. Horse was their coach, who could impose rules on what the athletes were allowed to eat. While the big dog was allowed to eat fried rice but not pork soup, the small dog was allowed to eat fried rice and he was allowed to eat pork soup. A puppet, Kermit, was asked to provide a reminder of the rules, and Kermit produced the test sentence in (7).

- (7) dagou keyi chi chaofan huozhe zhuroutang
 big.dog may eat fried.rice or pork.soup
 ‘The big dog is allowed to eat fried rice or pork soup.’

If participants computed the FCI, they were expected to reject the test sentence. Huang and Crain reported that children correctly rejected the target sentences 95% of the time, comparable to adults’ rejection rate of 98%. Huang and Crain took these results to corroborate the previous finding that children are able to license conjunctive inferences from disjunctive sentences containing a deontic modal verb.

Further evidence of children’s abilities with free choice comes from An et al. (2020), who investigated Mandarin-speaking children’s understanding of disjunction in the scope of the adverb *dou*, as in (8).

- (8) jiaozi huozhe shousi, gongfuxiongmao dou hui zuo
 dumplings or sushi KungFuPanda DOU can make
 Literal meaning: ‘Dumplings or sushi, Kung Fu Panda DOU can make.’
 Intended: ‘Kung Fu Panda can make dumplings and sushi.’

An et al. (2020) assume that the disjunctive phrase is interpreted in the scope of the adverbial quantifier *dou*, and is recursively exhaustified to give rise to a conjunctive FCI. In their Truth Value

Judgment Task, 4-year-old children (mean age 4.65) listened to stories and then had to judge a puppet’s description. In one target story, for example, Kung Fu Panda and his friends participated in a cooking school, and Kung Fu Panda stated that he could cook one of the two options between dumplings and sushi. The target sentence produced by the puppet was then the one in (8). If children generated a FCI, they were expected to reject the sentence. Indeed, children rejected such descriptions 87.5% of the time (comparable to adults’ rejection rate of 91.7%).

While the studies we have discussed up to this point have focused on studies of FCIs arising from the use of disjunction in deontic modal statements, it is worth noting the existence of acquisition studies of indefinites associated with FCIs, such as *any* in English and *renhe* in Mandarin, both of which can behave as a negative polarity item and as a free choice indefinite (e.g., *Kung Fu Panda is allowed to push any car*). While FC uses of such indefinites reportedly emerge later than NPI uses in spontaneous production (Tieu 2010, 2013), experimental studies using Truth Value Judgment Tasks reveal adult-like free choice interpretations of both *any* (Tieu et al. 2016, Experiment 2) and *renhe* (Huang and Crain 2014) in 4- and 5-year-old children. See also Huang et al. (2018) (Experiment 2) for a study of Mandarin-speaking children’s understanding of sentences containing the wh-word *shei* ‘who’ followed by *dou* ‘all’ and the modal verb *beiyunxu* ‘is allowed to’, such as (9). Here too, children are reported to display adult-like rates of FCIs.

- (9) Shei dou bei yunxu-le chi bingqilin
 who all PM allow-ASP eat ice-cream
 ‘Anyone was allowed to eat ice-cream.’

In the case of indefinites as well, then, children appear to be quite adult-like in accessing free choice interpretations.

Against the background of the preceding studies, which find consistently adult-like behavior on FCIs from children,² Cochard et al. (2025) (see also Cochard 2025) question the claim that children are genuinely adult-like in computing FCIs. Cochard et al. (2025) argue that children who were reported to compute adult-like FCIs from modal disjunctive statements might only have *appeared* to be adult-like. The authors argue that at least some of the children might have in fact interpreted the modal disjunctive statement as a modal *conjunctive* statement. The possibility of this conjunctive interpretation is related to the finding reported in various acquisition studies in the literature that children sometimes interpret disjunction as conjunction (Paris 1973; Braine and Romain 1981; Singh et al. 2016; Tieu et al. 2017; Skordos et al. 2020). That is, children are sometimes reported to interpret disjunctive sentences such as *Liz bought a croissant or a donut* conjunctively, as *Liz bought a croissant and a donut*. Cochard et al. (2025) suggest that if this conjunctive interpretation is also possible in modal contexts, it could explain at least some of the adult-like behavior observed in FC studies.

Cochard et al. (2025) note that in the critical trials in studies reporting adult-like FCI behavior, such as Tieu et al. (2016) and Huang and Crain (2020), the rules are set up in such a way that the relevant character is allowed to engage with one object but not another; the target disjunctive statement then mentions both objects, and is consequently false if interpreted with the FCI. As Cochard

²The majority of previous studies have focused on positive FCIs. See Huang et al. (2020) for a study of Mandarin-speaking children’s interpretation of modal disjunction in negative environments, the results of which indicate that negation cancels FCIs for children, who in an early stage of development, interpret the disjunction *huozhe* ‘or’ in the scope of negation, in contrast to adults, who treat the disjunction as polarity-sensitive and scope it above negation. In short, children at this stage appear to access the *double prohibition* reading of the negated disjunctive statements.

et al. (2025) point out, however, if children misunderstand the modal disjunction (*Liz can buy a croissant or a donut*) as a modal conjunction (*Liz can buy a croissant and a donut*), they would similarly be expected to reject the test sentences, giving the appearance of adult-like behavior, but for the wrong reasons. To address this issue, Cochard et al. present a new experiment with two key conditions designed to tease apart genuine FCIs from a modal conjunctive interpretation. A character would be at the shop with a certain number of coins in their purse, and could purchase certain items, depending on how much they cost. In the critical *Free Choice False* condition, Liz the Bear had exactly one coin; the croissant cost one coin (and thus Liz could afford it), while the donut cost two coins (and thus Liz would not be able to buy it). In such a scenario, the sentence in (10) would be false on both the free choice and modal conjunctive interpretations.

- (10) Avec une pièce, Liz l'ours peut acheter un croissant ou un donut.
 With one coin Liz the.bear can.3SG.PRS buy.INF a croissant or a donut
 'With one coin, Liz the bear can buy a croissant or a donut.'

This condition, similar to the critical conditions in previous studies, would not be sufficient to tease apart the two conjunctive interpretations. Cochard et al. (2025) thus included a *Mutually Exclusive* condition, in which Liz the bear again had one coin, but this time the croissant and the donut each cost one coin. In such a scenario, (10) would be true on a genuine free choice interpretation, since Liz was in principle able to buy the croissant and she was able to buy the donut, but the sentence would be false on a modal conjunctive reading, since she could not buy both at the same time. The authors found that children accepted the *Free Choice False* targets 27.2% of the time (significantly different from adults' 1.8% acceptance rate), while they accepted the *Mutually Exclusive* targets 76.2% of the time (compared to adults' 100% acceptance rate). Next, Cochard et al. (2025) analyzed individual participants' responses, and found that all adults except one consistently rejected the *Free Choice False* targets and accepted the *Mutually Exclusive* targets; the majority of children resembled adults in these two conditions, but there were nevertheless some children who consistently accepted the *Free Choice False* targets and children who consistently rejected the *Mutually Exclusive* targets. Cochard et al. thus conclude that there are subgroups of children, including those who are adult-like and genuinely access free choice, and those who are merely accessing a modal conjunctive interpretation.

In a study conducted with Romanian-speaking children, Bleotu et al. (2024, 2026) also investigated a possible link between children's conjunctive interpretations of plain disjunction and their ability to compute FCIs. To do so, the authors conducted two tasks with children, an *Implicature Task* testing children's interpretation of plain disjunctive sentences, and a *Free Choice Task* testing their interpretation of modal disjunctive sentences. Like Cochard et al. (2025), Bleotu et al. set up scenarios in such a way as to be able to distinguish genuine FC interpretations from modal conjunctive interpretations, and moreover attempted to investigate the hypothesis that children who interpret disjunction conjunctively should do so for both plain and modalized disjunctions. While some children did indeed exhibit conjunctive readings across both plain and modal disjunctions (consistent with the hypothesis advanced in Cochard et al. 2025), there were also subgroups of children who were conjunctive for plain disjunctions but not for modal disjunctions, and children who were conjunctive in the modal condition but not in the plain disjunction condition. The authors conclude that there may indeed be multiple routes to free choice behavior.

Finally, a recent study by Eliatamby (2025) reports lower rates of FCIs in children. Eliatamby

used a set-up in which the ‘rules’ were presented in a more abstract way. Participants were told that Larry’s mom was throwing out some of his things, and had issued the following rule: *Larry is (only) allowed to keep things with a red star on them*. In a critical context, participants would then see a picture containing a teddy, a train, and a ball, with a star only under the teddy. They then had to judge Larry’s utterance: *I’m allowed to keep the train or the teddy*. In this experimental set-up, Eliatamby reports 4- and 5-year-old children (mean age 5.05) computed significantly fewer FCIs compared to adults, accepting the critical targets 59–64% of the time. According to Eliatamby, the reason previous studies found higher rates of FCIs in children is that the disjunct alternatives required to compute the FCI, for example the green and orange cars in the Tieu et al. (2016) study, were made contextually available (and quite salient) in the test stories – similarly to how contextual alternatives are easily available to children when successfully computing ad hoc implicatures (Stiller et al. 2015).

In sum, there have been multiple studies of children’s ability to access FCIs, in different constructions, across different languages (notably Mandarin, English, French, and Romanian), and while there is some amount of variation in the findings, the general consensus seems to be that FCIs are mastered earlier than standard scalar implicatures. This asymmetry poses a challenge for implicature-based accounts of FCIs.³

This result has been interpreted in two main ways. One line of response takes it as support for non-implicature accounts of FCIs (see, for example, Aloni 2022 and Goldstein 2019). Another maintains the implicature-based approach but posits that the ingredients involved in deriving FCIs differ in important ways from those involved in deriving standard SIs. An idea advanced in Chemla and Bott (2014) and Tieu et al. (2016) is that the asymmetry between the two kinds of inferences is related to the nature of the alternatives that are required. In the case of FCIs, the required alternatives correspond to the individual disjunct alternatives, which can be obtained through simplification (see Katzir 2007); in the case of standard SIs typically tested in developmental and adult processing studies, the required alternatives are obtained through lexical replacement. As we will see, however, this line of explanation faces difficulties in accounting for the second challenge, to which we now turn.

2.2.2 The polarity asymmetry

The second challenge comes from comparing the robustness of FCIs from sentences such as (1) with that of their negative counterpart in (3), namely *negative free choice* (Fox 2007).

The implicature approach derives negative FCIs in the same way as it derives FCIs, on the basis of alternatives paraphrasable as in (11), the negation of which conveys that Mia does not have to buy the apple and she does not have to buy the pear.

- (11) a. Mia only doesn’t have to buy the apple.
 b. Mia only doesn’t have to buy the pear.

The prediction of the implicature approach here is again straightforward: FCIs and NFCIs are derived in the very same way, on the basis of the same type of alternatives, so, all else being equal,

³A related challenge comes from adult processing. If FCIs and scalar implicatures share a common derivation, they might be expected to exhibit similar processing profiles. However, Chemla and Bott (2014) provide experimental evidence that FCIs, unlike standard scalar implicatures, do not give rise to a delay in processing.

they should be computed by adults to a similar extent; that is, they should be similarly robust.

Again, this is a prediction that has been challenged in the literature (Marty et al. 2021; Marty et al. 2024a). The main finding has been that while FCIs are a very robust inference, almost always associated with sentences like (1), their negative counterparts are much weaker. Importantly, Marty et al. (2021) compared FCIs and NFCIs to positive and negative scalar implicatures in order to control for the role of negation as well as for potential differences between weak and strong scale-mates. That is, they compared FCIs to positive SIs such as (2) and NFCIs to negative SIs such as (4). Their results show that, in contrast to NFCIs, the negative implicature in (4), if anything, was more robust than its positive counterpart. This result constitutes a clear challenge to the implicature approach to FCIs, and also to the response based on alternatives mentioned above, given that FCIs and NFCIs are based on the same kind of alternatives, i.e. simplified versions of the asserted sentence.

Finally, Marty et al. (2024a) extend these results to different modal flavors and related inferences. They also investigated sentences such as (12), where negation is placed below the modal, creating an equivalence with (3) while reintroducing disjunction rather than conjunction. Such a structure would allow them to test whether the crucial factor underlying the difference between FCIs and NFCIs is the negation or rather the distinction between conjunction and disjunction. Their results reveal that *low NFCIs*, as they call them, are as robust as positive FCIs, suggesting that disjunction, rather than negation, might be the relevant factor.

- (12) Mia is allowed not to buy the apple or not to buy the pear.
 $\rightsquigarrow$ *Mia is allowed not to buy the apple and she is allowed not to buy the pear*

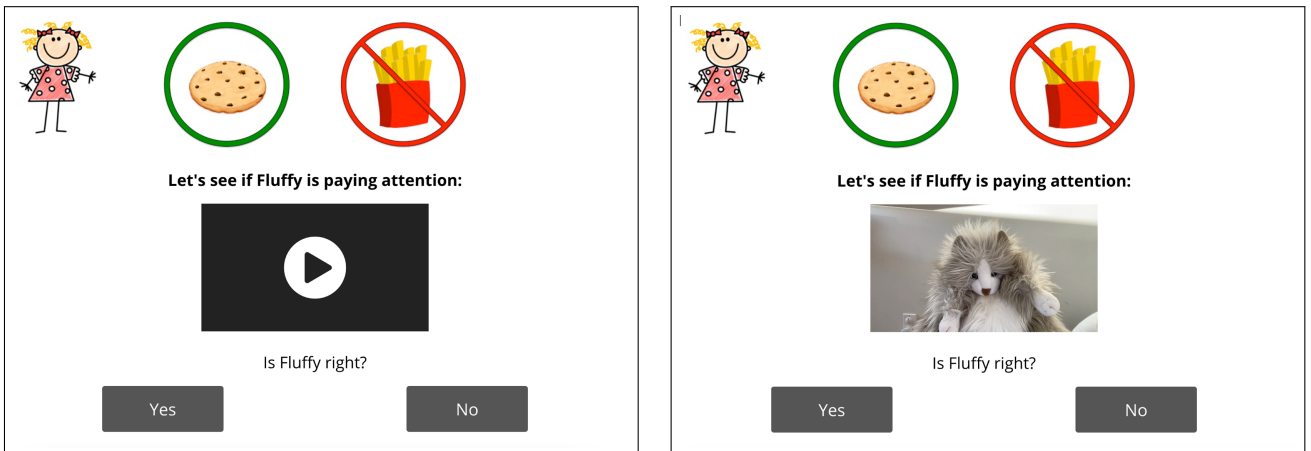
2.3 The response: the hybrid approach

As discussed, the implicature approach has faced two main challenges in recent years. While an alternatives-based response may address the first (see Tieu et al. 2016; but see Marty et al. 2021 for a critical assessment), the second remains unresolved. A different line of explanation is developed in Marty et al. (2021) and Marty et al. (2024a) in the form of a hybrid approach. The core idea is that non-implicature accounts may be correct for positive FCIs, whereas the implicature approach is correct for NFCIs. Although this proposal is less parsimonious, it alleviates both challenges. First, if FCIs are not derived via implicature, there is no longer any expectation that children should behave with FCIs as they do with standard implicatures. Second, if NFCIs and FCIs have distinct sources, there is no reason to expect them to match in robustness.

The hybrid approach not only addresses these challenges but also makes a clear further prediction (as noted by Marty et al. 2021 and Marty et al. 2024a). From an acquisition perspective, the locus of parallelism shifts from FCIs to NFCIs: if NFCIs are standard scalar implicatures, then children should acquire them in a similar way and master them on a comparable timeline. More specifically, children should behave with NFCIs as they do with negative SIs such as (4). This is the main prediction tested in the present study, to which we now turn.

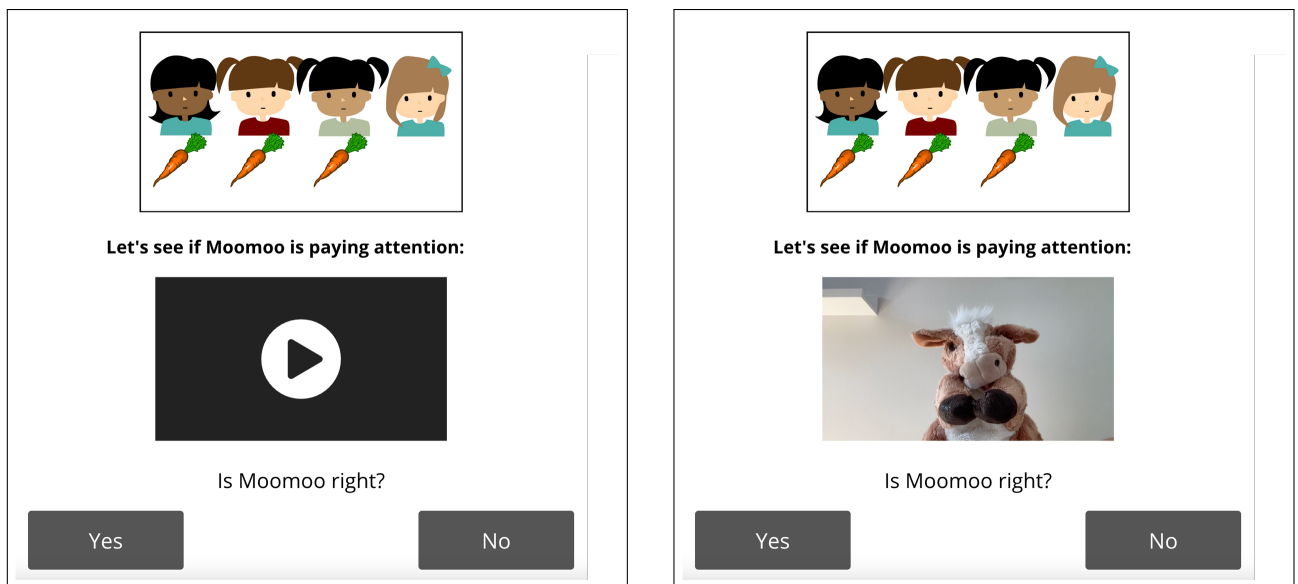
A. Experiment 1: Modal quantifiers

Example target item for the positive FC sentence *Mia is allowed to buy either the cookie or the fries.*



B. Experiment 2: Nominal quantifiers

Example target item for the positive FC sentence *Some girls have either a carrot or a mushroom.*



(a) Item onset

(b) Video playback

Figure 1: Example items from Experiment 1 (modal quantifiers; top) and Experiment 2 (nominal quantifiers; bottom), illustrating the visual display and trial procedure. The left panel shows the item at onset and the right panel shows what would appear during video playback of the puppet's utterance.

3 Experiment 1

Experiment 1 tests the predictions of the hybrid approach by comparing positive and negative FCIs in the modal domain with their positive and negative SI counterparts.

3.1 Methods

3.1.1 Participants

We tested 34 children (5;02–7;03, $M=6;03$) at a school in Toronto, Canada. Children were tested on an iPad or laptop computer in a quiet room at their school. We also recruited 35 English native speaker adult controls through Prolific; all indicated English as their first language, were residing in the USA, Canada, or the UK, had normal or corrected-to-normal vision, and had a previous Prolific approval rating of 99-100%. Adult participants were paid at an average hourly rate of £16.04 for their participation.

3.1.2 Materials

Each experimental item displayed a character alongside two food items, with a video player shown below (see Figure 1A for an illustration). The character was one of two children, *Sam* or *Mia*.⁴ The two food items were distinct and commonly available products. Each food item was enclosed in one of three symbols: a green circle, a red circle with a diagonal line, or a black square.⁵ Participants were instructed to interpret these symbols as indicating what the two children were *allowed to buy* (green circle), *not allowed to buy* (red circle), and *required to buy* (black square) at the store, according to their parents (see Appendix A for details). Pairwise combinations of the three symbols, allowing repetition and disregarding order, yielded the six picture types illustrated in Table 1.

The video player was configured to play, upon clicking, a short clip featuring a puppet named Fluffy. In each clip, the puppet uttered a sentence pertaining to the depicted situation. All sentences were constructed such that the puppet was always correct about the identity of the character and the type of food items involved. However, the puppet could be incorrect or underinformative with respect to what the character could or could not buy, and what they were required or not required to buy. On screen, the video player was introduced by the anchor sentence “Let’s see if Fluffy is paying attention” and followed by the probe question “Is Fluffy right?”.

Test sentences were positive and negative sentences which could yield free choice and scalar inferences. They were constructed according to the four sentence frames illustrated in Table 2. FC sentences were formed by embedding a disjunction under the possibility modal *allowed to* in the positive condition (PosFC), and a conjunction under the negated necessity modal *have to* in the negative condition (NegFC). Positive and negative SI sentences were obtained from their FC counterparts by simplification: by removing one of the disjuncts in the positive condition (PosSI) and one of the conjuncts in the negative one (NegSI).

FC and SI sentences were paired with different picture types to create, for each sentence type, TARGET items probing the inference of interest, along with TRUE and FALSE controls. Specifically, in TARGET items, sentences were paired with picture types that would make them false if the inference of interest were present, but true if it were absent. In these items, for instance, PosFC sentences were paired with AD pictures, in which one of the disjuncts was represented as ‘allowed’

⁴The identity of the character was pseudo-randomized such that each character appeared in half of the items.

⁵The study used a set of 18 food items. For consistency, each item was always paired with the same symbol across trials. The set was divided evenly among symbols: six items were associated with the green circle (*pear, broccoli, pineapple, cookie, popsicle, croissant*), six with the red circle (*burger, hot-dog, cupcake, fries, pizza, donut*), and six with the black square (*banana, carrot, apple, avocado, orange, lettuce*).

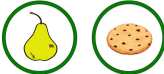
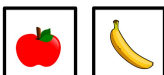
Picture type	Description of the picture	Example picture
DD	Disallowed to buy either item	
AD	Allowed to buy one item & Disallowed to buy the other	
AA	Allowed to buy both items	
AR	Allowed to buy one item & Required to buy the other	
DR	Disallowed to buy one item & Required to buy the other	
RR	Required to buy both items	

Table 1: Description and example of the picture types used in Experiment 1. The position of the food items on the screen (on the right vs. on the left) was randomized.

and the other as ‘disallowed’. In control items, by contrast, sentences were paired with picture types that made them unambiguously true or false, regardless of whether the target inference was present. A summary of the sentence–picture combinations used to create these items is provided in Table 3, where *L* and *R* serve as cover terms for the food items displayed on the left and right of the example pictures, respectively. For each test sentence, there were two TRUE, two FALSE, and four TARGET items, yielding 16 control and 16 target items in total.

The control set further included Have-To sentences featuring the necessity modal *have to* in a positive environment, of the form ‘[NAME] has to buy the [L/R]’ (e.g., *Mia has to buy the banana*). These sentence were paired with pictures in which the relevant food item was represented as either ‘required’ (black square) or ‘allowed’ (green circle), making the sentences true in the former case and false in the latter. These items served to further verify that participants understood the distinction between the green circle (allowed but not required) and the black square (required). Four such items were included, two TRUE and two FALSE, bringing the total number of items in the experiment to 36.

3.1.3 Procedure

In the instructions, participants were introduced to *Fluffy*, a friendly puppet, and two child characters, *Sam* and *Mia*. They were told that Sam and Mia were shopping at a store and that their parents had set rules specifying what they were allowed to buy, not allowed to buy, and required to buy (see Appendix A). Following the set-up in Marty et al. (2021, 2024a) (see also Tieu et al. 2024), participants were introduced to three symbols representing these rules: a green circle around an item indicated the character was allowed to buy that item, a red circle with a line through an item

Sentence type		
PosFC	Frame:	[NAME] is allowed to buy either the [L] or the [R].
	Example:	‘Mia is allowed to buy either the pear or the burger’.
	Inference:	$\leadsto$ <i>Mia is allowed to buy one and allowed to buy the other</i>
NegFC	Frame:	[NAME] doesn’t have to buy both the [L] and the [R].
	Example:	‘Sam doesn’t have to buy both the donut and the banana’.
	Inference:	$\leadsto$ <i>Sam isn’t required to buy one and isn’t required to buy the other</i>
PosSI	Frame:	[NAME] is allowed to buy the [L/R].
	Example:	‘Sam is allowed to buy the banana’.
	Inference:	$\leadsto$ <i>Sam isn’t required to buy the banana</i>
NegSI	Frame:	[NAME] doesn’t have to buy the [L/R].
	Example:	‘Mia doesn’t have to buy the donut’.
	Inference:	$\leadsto$ <i>Mia is allowed to buy the donut</i>

Table 2: Sentence frames used in Experiment 1, illustrated with examples and the free choice and scalar inferences of interest. [NAME] was the name of the relevant character; [L] and [R] were the names of the food items displayed on the left and right of the corresponding pictures, respectively. For sentences involving only one food term (PosSI and NegSI), the food name was pseudo-randomly assigned to match one of the two displayed items (either left or right).

indicated the character was not allowed to buy that item, and a black square around an item indicated the character was required to buy that item. Participants were told that they would see pictures depicting these rules and that, in each trial, Fluffy would say what Sam or Mia could or could not buy according to the rules. The participant’s task was to decide whether Fluffy was right or wrong.

Following the instructions, participants completed six practice trials designed to familiarize them with the task and reinforce their understanding of the symbols. These trials were visually identical to the experimental trials. Practice items involved PosSI, NegSI and Have-to sentences (i.e., *Mia {is allowed to, doesn’t have to, has to} buy the [food].*), each presented once with a picture that made it clearly true and once with a picture that made it clearly false. After each response, participants received brief feedback explaining the correct answer, e.g., *In this case, you should say Yes/No, because Fluffy is right/wrong: Mia DOES/DOESN’T have to buy the [food].* After the practice phase, participants were informed that no further feedback would be provided and proceeded to the experimental trials.

In both the practice and test phases, items were presented in random order with a 500 ms interstimulus interval. The probe question, *Is Fluffy right?*, was repeated on each trial. Adult participants could replay the video clip by clicking on it and responded by selecting one of two buttons labelled ‘Yes’ and ‘No’. Child participants provided their responses orally to the experimenter, who recorded them by selecting the corresponding response button.

	FALSE	TARGET	TRUE
PosFC	 	 	 
Mia is allowed to buy either the <i>L</i> or the <i>R</i> .	DD (2)	AD (4)	AA (2)
NegFC	 	 	 
Mia doesn't have to buy both the <i>L</i> and the <i>R</i> .	RR (2)	AR (4)	AA (2)
PosSI	 	 	 
Mia is allowed to buy the <i>L</i> .	DD (2)	DR (4)	AD (2)
NegSI	 	 	 
Mia doesn't have to buy the <i>L</i> .	RR (2)	DR (4)	AR (2)

Table 3: Sentence–picture combinations used to construct the FALSE, TARGET, and TRUE items in Experiment 1. Each combination is illustrated with a schematic example where *L* refers to the food item on the left and *R* to the one on the right. Numbers in parentheses indicate the number of items per condition.

3.1.4 Participant exclusion criteria

Participants who failed to reach (a pre-determined threshold of) 75% accuracy on control trials (15/20 correct responses), corresponding to above-chance performance in a binomial test against chance level ($p = .5$, $p < .05$), were excluded prior to analysis. This resulted in the removal of three participants (two adults and one child), yielding a final sample of 33 adults and 33 children.

3.2 Results

Mean rates of ‘No’ responses (i.e., mean rejection rates) to the test sentences are shown in Figure 2.

Control items Responses to the control items were as expected. Both groups strongly rejected the test sentences in their FALSE condition (all mean rejection rates $\geq 96\%$) and accepted them in their TRUE condition (all rejection rates $\leq 8\%$).⁶ Overall accuracy was high, reaching 96% [94, 97] for children and 97% [96, 98] for adults. These results indicate that both adults and children performed the task appropriately and, crucially, had no difficulty mapping symbols to their intended meanings and interpreting the test sentences relative to the situations they depicted.

Target items Responses to the TARGET items were first analyzed using separate mixed-effects logistic regression models for adults and children. In both cases, the dependent variable was *Response* (coded as 1 for ‘No’ and 0 for ‘Yes’), modeled as a function of *Inference* (SI vs. FC), *Polarity* (Negative vs. Positive), and their interaction. Both predictors were sum-coded, and models included by-participant random intercepts and random slopes for *Inference* and *Polarity*.⁷ To assess the contribution of the interaction term, we compared each full model to a reduced model

⁶Here and in the following, values in square brackets indicate 95% binomial (Wilson) confidence intervals.

⁷Model structure: `Response ~ Inference * Polarity + (1 + Inference + Polarity | Subject)`

differing only in the absence of the interaction, using likelihood-ratio tests. Estimated marginal means were used for planned pairwise comparisons between levels of Polarity within each level of Inference, as well as for post-hoc analyses.

In both groups, the interaction between *Inference* and *Polarity* was significant (Adults: $\beta = 0.32$, $z = 4.68$, $p < .001$; Children: $\beta = 0.43$, $z = 6.40$, $p < .001$) and improved model fit (Adults: $\chi^2(1) = 22.11$, $p < .001$; Children: $\chi^2(1) = 41.96$, $p < .001$). This interaction reflects broadly similar, though not identical, patterns across groups. In adults, there were clear polarity effects going in opposite directions for FC and SI: adults rejected FC sentences less often in their negative than in their positive form (66% [57, 73] vs. 98% [94, 100]; $z = -4.00$, $p < .001$), whereas they rejected SI sentences more often in their negative than in their positive form (37% [29, 46] vs. 17% [12, 25]; $z = 2.22$, $p = .026$). In children, the polarity contrast for FC was much smaller (Negative: 86% [79, 91]; Positive: 98% [94, 99]; $z = -1.93$, $p = .052$), whereas the contrast for SI was markedly larger, with much higher rejection rates in the negative than in the positive cases (89% [82, 93] vs. 26% [19, 34]; $z = 7.15$, $p < .001$).

To better understand the difference between adults and children in negative FC and SI trials, we conducted a post-hoc analysis focusing on these in the FALSE and TARGET conditions. We fitted a combined model including *Inference*, *Condition* (False vs. Target), and *Group* (Adults vs. Children), with all interactions.⁸ This model revealed significant interactions between *Inference* and *Condition* ($\beta = 0.75$, $z = 3.13$, $p = .002$) and between *Condition* and *Group* ($\beta = 0.53$, $z = 2.15$, $p = .031$), while the three-way interaction was not significant ($p = .65$). Post-hoc pairwise comparisons showed that, for SI, rejection rates were higher in the FALSE than in the TARGET condition in both groups. For FC, however, this contrast was significant in adults but not in children. In other words, unlike adults, children rejected negative FC sentences at similarly high rates in the FALSE and TARGET conditions.

3.3 Summary

The present data reproduce the polarity contrasts observed in Marty et al. (2021) for adults: FC-related sentences elicited more FC-type responses in their positive than in their negative forms, whereas SI-related sentences elicited slightly more SI-type responses in their negative than in their positive forms. This pattern, however, only partially extends to children. While children showed a clear polarity contrast for SI in the same direction as adults, albeit much stronger, there was no comparable contrast for FC. Specifically, negative FC sentences, like their positive counterparts, were rejected at similarly high rates in both the FALSE condition, where they were literally false—and the TARGET condition—where they were true on their non-FC reading but false on their FC reading.

Taken together, the results do not provide straightforward support for hybrid accounts. On the one hand, such accounts correctly capture the contrast between FC and SI in positive sentences, as well as the parallel behavior of FC and SI in negative sentences. On the other hand, the developmental pattern departs in important ways from their predictions. In particular, children do not exhibit the expected asymmetry between positive and negative FC, and instead show uniformly high rejection rates for positive and negative FC. While hybrid approaches are well suited to capture polarity asymmetries, they do not directly predict this pattern, nor do they offer an immediate

⁸Model structure: (Rejection~Inference*Condition*Group+(1+Condition+Inference|Subject))

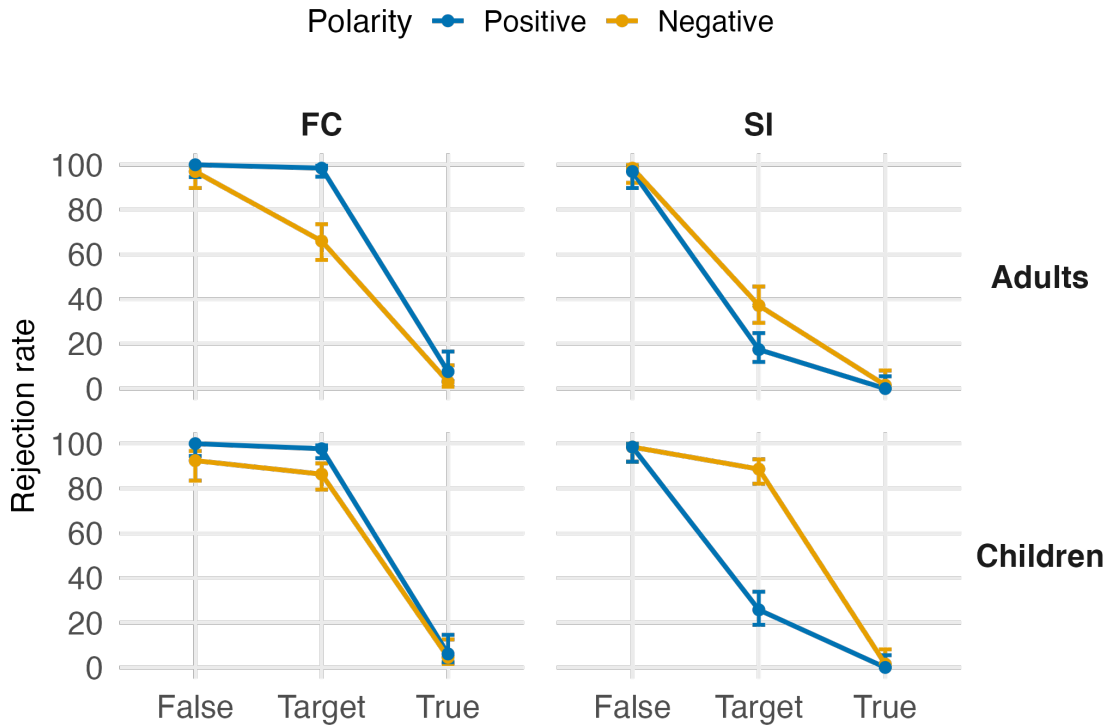


Figure 2: Mean rates of ‘No’-responses across conditions, by inference type, polarity, and participant group. In the TARGET conditions, this measure serves as a proxy for the rate of with-FC/SI responses. Error bars indicate 95% binomial (Wilson) confidence intervals.

explanation for the elevated rejection rates observed in children.

4 Experiment 2

Experiment 2 extends the investigation to the nominal quantifier domain in order to test whether the polarity asymmetry observed in adults also arises in this domain, and to further examine the predictions of the hybrid approach. In addition, Experiment 2 includes a variant of the negative FC condition in which negation appears lower and conjunction is replaced with disjunction, allowing us to assess whether the contrast between positive and negative FC sentences is driven by the presence of negation itself or by the type of connective involved.

4.1 Methods

4.1.1 Participants

We tested 34 children (5;05–7;04, $M=6;05$) at a school in Toronto, Canada.⁹ As in Experiment 1, children were tested on an iPad or laptop computer in a quiet room at their school. We also recruited 36 English native speaker adult controls through Prolific; all indicated English as their first language, were residing in the USA, Canada, or the UK, had normal or corrected-to-normal vision, had a Prolific approval rating of 99–100%, and had not participated in Experiment 1. Adult participants were paid at an average hourly rate of £11.80 for their participation.

4.1.2 Materials

The item display and trial procedure were the same as in Experiment 1. Each experimental item consisted of a picture with a video player displayed below it (see Figure 1B for an illustration). All pictures depicted four horizontally aligned child characters—either all boys or all girls—each appearing with either one object, two objects, or none.¹⁰ The video player was configured, as before, to play a short clip of a puppet (here, *Moomoo*) uttering a sentence describing the situation depicted above. The video player was introduced by the same anchor sentence (i.e., “Let’s see if Moomoo is paying attention”) and followed by the same probe question (i.e., “Is Moomoo right?”) as in Experiment 1.

Test sentences were constructed according to the frames illustrated in Table 4. FC sentences were non-modalized quantificational sentences which, under an implicature-based approach, are expected to yield free choice inferences comparable to those observed with their modal counterparts. Three types of FC sentences were included: PosFC sentences, formed by embedding positive disjuncts under the existential quantifier *some*; LowNegFC sentences, formed by embedding negated disjuncts under *some*; and HighNegFC sentences, formed by embedding positive conjuncts under the universal quantifier *all* within the scope of negation. SI sentences were modeled after the PosFC and HighNegFC sentences, yielding the corresponding PosSI and HighNegSI sentences.¹¹

As in Experiment 1, FC and SI sentences were paired with different picture types to create, for each sentence type, TARGET items probing the inference of interest, along with TRUE and FALSE control items. Crucially, FC sentences (Pos, LowNeg and HighNeg) were always paired with pictures that made true the potential SI arising from the more general competition between *some* and *all* (e.g., *Some girls have either a [A] or a [B]* $\rightsquigarrow$ *Not all girls have a [A] or a [B]*). A summary of the sentence–picture combinations used to construct these items is provided in Table 5. In this table, pictures are schematically described with respect to the objects shown in front of each character: *A* and *B* correspond to objects mentioned in the sentence, *C* and *D* to

⁹All children had previously participated in Experiment 1. We reasoned that this was acceptable given that the visual and linguistic materials of the two experiments were entirely different, and the puppet that “interacted with” participants through the pre-recorded videos was also different. The two testing sessions generally took place about two to three months apart.

¹⁰Half of the items involved boys and the other half girls.

¹¹Another negative SI variant can be formed by embedding negation under *some*, as in *Some boys don’t have a coconut*, paralleling the LowNegFC sentences. Such sentences are expected to yield a lower-bounding ‘existential’ inference, similar to their high-negation counterparts (i.e., *Some boys have a coconut*). These variants were not included in the present study; see Ramotowska et al. (2024) for a comparison across these cases in adult comprehension, as well as evidence that *some-not* patterns like *not-all* in yielding higher rates of with-SI responses than *some*.

Sentence type		
PosFC	Frame:	Some {girls, boys} have either a [A] or a [B].
	Example:	“Some boys have either a kiwi or an orange.”
	Inference:	$\rightsquigarrow$ <i>Some boys have a kiwi and some have an orange</i>
LowNegFC	Frame:	Some {girls, boys} either don’t have a [A] or don’t have a [B].
	Example:	“Some girls either don’t have a cookie or don’t have a kiwi.”
	Inference:	$\rightsquigarrow$ <i>Some girls don’t have a cookie and some don’t have a kiwi</i>
HighNegFC	Frame:	Not all {girls, boys} have both a [A] and a [B].
	Example:	“Not all boys have both a pencil and a dinosaur.”
	Inference:	$\rightsquigarrow$ <i>Not all boys have a pencil and not all of them have a dinosaur</i>
PosSI	Frame:	Some {girls, boys} have a [A].
	Example:	“Some girls have a pear.”
	Inference:	$\rightsquigarrow$ <i>Not all girls have a pear</i>
HighNegSI	Frame:	Not all {girls, boys} have a [A].
	Example:	“Not all boys have a coconut.”
	Inference:	$\rightsquigarrow$ <i>Some boys have a coconut</i>

Table 4: Sentence frames used in Experiment 2, illustrated with examples and their expected free choice and scalar inferences. The subject NP (*boys* or *girls*) varied across trials. [A] and [B] are placeholders for singular nouns.

other (unmentioned) objects, and *X* indicates that no object was present. For example, in the TARGET items, PosFC sentences were paired with A–A–A–X pictures, in which three characters each had an *A* (and nothing else) and one character had no object. This configuration makes the sentence false under the free choice interpretation (since no character has a *B*), but true otherwise, irrespective of whether the potential *not-all* SI is derived.

4.1.3 Procedure

The procedure was similar to that used in Experiment 1 (see 3.1.3). In the instructions, participants were introduced to a friendly puppet, *Moomoo*, who appeared in pre-recorded videos under the guise of interacting with participants. Participants were told that they would see pictures of boys and girls and the things they have, and that Moomoo would describe each picture (see Appendix B). Their task was to decide whether Moomoo was right or wrong.

Before the test phase, participants completed four practice trials designed to familiarize them with the task. Practice items were visually similar to the experimental items but involved simpler displays depicting a single character (a boy or a girl) with a single object. Sentences in these items were simple positive and negative statements with a singular definite subject (i.e., *The {girl, boy} has/doesn’t have a(n) [object]*), each presented once with a picture that made it clearly true and once with a picture that made it clearly false. Positive sentences were presented first, followed by negative ones. After each response, participants received brief feedback explaining the correct

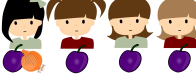
	FALSE	TARGET	TRUE
PosFC Some girls have either a [A] or a [B].	 C-D-C-X (2)	 A-A-A-X (4)	 A-B-A-X (2)
LowNegFC Some girls either don't have a [A] or don't have a [B].	 AB-AB-AB-AB (2)	 AB-A-A-A (4)	 AB-A-B-A (2)
HighNegFC Not all girls have both a [A] and a [B].	 AB-AB-AB-AB (2)	 AB-A-A-A (4)	 AB-A-B-A (2)
PosSI Some girls have a [A].	 C-D-C-X (2)	 A-A-A-A (4)	 A-A-C-X (2)
HighNegSI Not all girls have a [A].	 A-A-A-A (2)	 C-C-C-X (4)	 A-B-A-X (2)

Table 5: Sentence–picture combinations used to construct the FALSE, TARGET, and TRUE items in Experiment 2. Pictures are schematically described in reference to the objects shown in front of each character: *A* and *B* correspond to objects mentioned in the sentence, *C* and *D* to (other) unmentioned objects, and *X* indicates that no object was present. Numbers in parentheses indicate the number of items per condition.

answer (e.g., *In this case, Moomoo is right/wrong: the girl DOES/DOESN'T have a(n) [object]*). After the practice phase, participants were informed that no further feedback would be provided and proceeded to the experimental trials.

In the test phase, experimental items were presented in random order. The probe question, “Is Moomoo right?”, was repeated on each trial. As in Experiment 1, adult participants could replay the video clip by clicking on it, and provided their responses by selecting one of two buttons labelled ‘Yes’ and ‘No’. Child participants provided their responses orally to the experimenter, who recorded them by selecting the corresponding response button.

4.1.4 Participant exclusion criteria

As in Experiment 1, participants who failed to reach the pre-determined accuracy threshold on control trials (75%, i.e., 15/20 correct responses) were to be excluded from the analysis. All

participants met this criterion, and thus none were excluded.

4.2 Results

Mean rates of ‘No’ responses (i.e., mean rejection rates) are shown in Figure 3.

Control items Responses to the SI controls were as expected, with an overall accuracy of 96% in both groups, indicating that both adults and children performed the task appropriately. Accuracy for the FC controls was lower (adults: 93%, children: 81%), reflecting higher-than-expected rejection rates in the TRUE conditions.

Specifically, in adults, both positive and low-negation FC sentences yielded rejection rates of around 17% in the TRUE conditions, higher than those observed for high-negation FC (2%) and for positive and negative SI sentences (4% and 0%, respectively). Children showed a more pronounced pattern, with rejection rates in these conditions ranging from 20% (positive FC) to 51% (low-negation FC). Since positive FC sentences showed this behavior in both groups, this discrepancy is unlikely to be solely driven by processing difficulties associated with negation. Rather, it suggests that other interpretive aspects of the FC stimuli, which we did not control for, contributed to increased rejection in the TRUE conditions.

One possibility, which we develop further in the Discussion, is that participants were sensitive to multiplicity inferences arising from the basic FC and inclusivity inferences associated with the FC sentences, e.g., the inference that *more than one girl* {has/doesn’t have} *a(n) [A] and more than one girl* {has/doesn’t have} *a [B]* for PosFC and LowNegFC sentences, and that *more than one girl has both a(n) [A] and a [B]* for HighNegFC sentences. Crucially, these additional inferences—false in our TRUE conditions—may arise only if the corresponding FC and inclusivity inferences are present in the first place. More importantly for our purposes, the resulting interpretations are likewise false in the TARGET conditions, and therefore do not affect the comparisons of interest.¹²

Target items Responses to the TARGET items were analyzed using the same pipeline as Experiment 1, namely separate mixed-effects logistic regression models for adults and children. We first restricted the analysis to the Positive and High Negation cases, the nominal counterparts of the modal cases tested in Experiment 1. In both models, the dependent variable was *Response*, predicted by *Inference* (SI vs. FC; sum-coded), *Polarity* (Positive vs. High Negation; sum-coded), and their interaction. The random-effects structure was identical to that used previously. In both groups, the interaction between *Inference* and *Polarity* was significant (Adults: $\beta = 0.41$, $z = 6.30$, $p < .001$; Children: $\beta = 0.39$, $z = 5.96$, $p < .001$) and improved model fit (Adults: $\chi^2(1) = 40.79$, $p < .001$; Children: $\chi^2(1) = 36.40$, $p < .001$), indicating a comparable response pattern across groups.

First, positive and negative SI sentences were rejected to a similar extent within each group. Children, however, showed overall higher rejection rates for SI sentences than adults, as well as a numerical trend toward higher rejection of negative SI sentences (Adults: 44% [36, 52] vs. 40% [32, 48], $z = 0.41$, $p = .67$; Children: 72% [64, 79] vs. 89% [83, 93], $z = 1.55$, $p = .12$). Second,

¹²For completeness, we carried out a secondary exploratory analysis of the present data using more stringent inclusion criteria, requiring accuracy above 75% for both inference types. This resulted in 7 exclusions, leaving 34 adults and 27 children. The overall pattern of results was virtually unchanged.

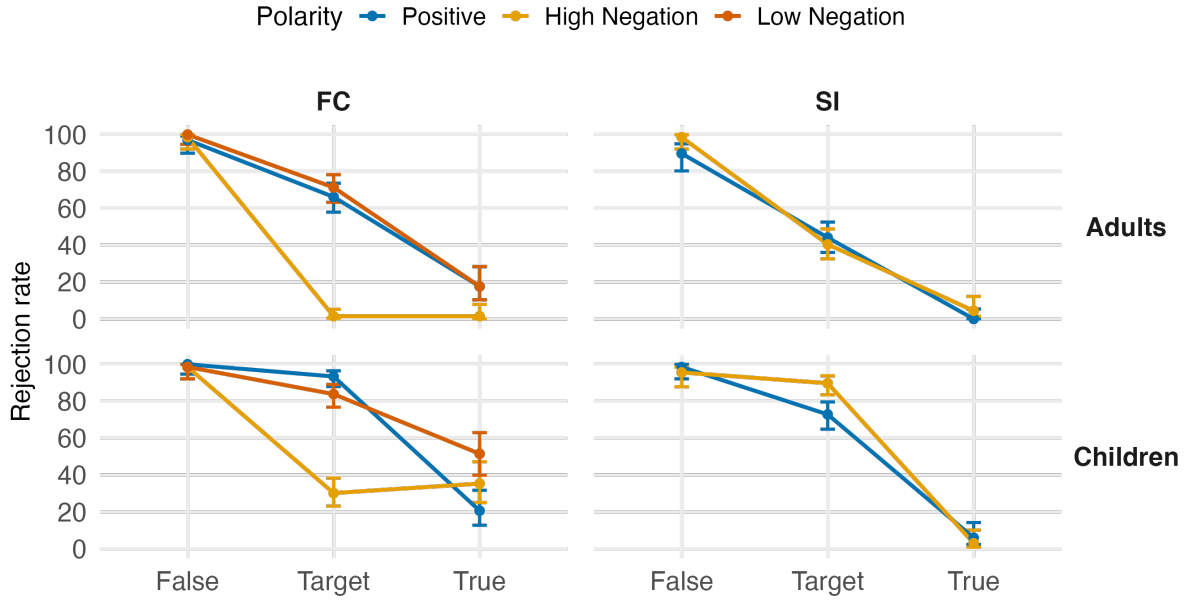


Figure 3: Mean ‘No’ response rates across conditions, by inference type, polarity, and participant group. Error bars indicate 95% binomial (Wilson) confidence intervals.

in both groups, FC sentences were rejected significantly less often in their (high) negative than in their positive form (Adults: 1.4% [0.4, 5.2] vs. 66% [57, 73], $z = -7.86$, $p < .001$; Children: 30% [23,38] vs. 93% [87,96]; $z = -6.4$, $p < .001$). While this polarity contrast parallels that observed in Experiment 1, rejection rates for (high) negative FC sentences in the TARGET conditions were comparable to those observed in the TRUE baselines within each group (Adults: 1.4% [0.07,7.8]; Children: 35% [25, 47]). Therefore, the present results provide no evidence for the existence of negative free choice inferences in the nominal domain.

Next, responses to FC sentences in the TARGET conditions were analyzed using a mixed-effects logistic regression model predicting *Response* from *Polarity* (levels: Positive, High negation, Low negation; dummy-coded), *Group* (Adults vs. Children; sum-coded), and their interaction.¹³ Estimated marginal means were then used to test pairwise contrasts between the three FC sentence types within each group.¹⁴ In both groups, Low-Negation FC yielded high rejection rates (Adults: 71% [63,78]; Children: 83% [76,89]), comparable to those for Positive FC (Adults: $z = -1.02$, $p = .55$; Children: $z = 1.65$, $p = .22$), and significantly higher than those for High-Negation FC (Adults: $z = -5.64$, $p < .001$; Children: $z = -4.98$, $p < .001$).

4.3 Summary

Experiment 2 was designed to extend our investigation of FCIs and NFCIs to the nominal quantifier domain. Specifically, we tested whether the polarity, and more generally the disjunction-conjunction asymmetry effects reported by Marty et al. (2024a) extended to the nominal domain,

¹³Model structure: $\text{Rejection} \sim \text{Polarity} * \text{Group} + (1 + \text{Polarity} + \text{Group} | \text{Subject})$

¹⁴ p -values were adjusted using the Tukey method to account for multiple comparisons within each group.

and whether the predictions of the hybrid approach would likewise be borne out in this domain.

At a general level, our results are consistent with the predictions of the hybrid approach. First, in both adults and children, FC sentences elicited substantially more FC-type responses in their positive than in their high-negation forms, replicating the pattern from Experiment 1. Second, disjunctive FC sentences embedding negation patterned with their positive counterparts, yielding high rates of with-FC responses, well above those observed for their conjunctive equivalents. These findings extend previous results from Marty et al. (2024a) and suggest that, as in the modal domain, FC sentences with nominal quantifiers readily give rise to FCIs when disjunction is involved, regardless of the presence of negation.

Importantly, however, the present data provide no evidence for an analogue of NFCIs in the nominal domain: in the TARGET conditions, the corresponding high-negation forms involving conjunction yielded low rejection rates, comparable to those observed in the TRUE baselines within each group. By contrast, high-negation SI sentences yielded rejection rates at least as high as their positive counterparts and far above those observed in the relevant TRUE baselines, in line with the results of Experiment 1. Taken at face value, this discrepancy between negative SI and negative FC poses a challenge for the hybrid approach: if NFCIs are derived as scalar implicatures, they should also arise in the nominal domain and pattern with negative SIs, as observed for their modal counterparts in Experiment 1. This expectation is not borne out by the results of Experiment 2.

5 Discussion

5.1 Main findings and theoretical implications

Our results first replicate the two main challenges to the implicature approach with which we began. Children are adult-like with positive FCIs, and adults derive positive FCIs more robustly than their negative counterparts, NFCIs. In addition, we replicate the result reported by Marty et al. (2024a) that low FC is as robust as positive FCIs despite the presence of negation. Finally, we extend the findings of Marty et al. (2021) to the nominal domain, showing that here too, for adults, positive FCIs are much more robust than NFCIs. Indeed, in this domain NFCIs are so weak that we find no evidence for them at all.

These findings thus provide both replications and extensions of the existing literature. Turning to the central test of the hybrid approach, however, our conclusion is more critical. The key developmental prediction of the hybrid account is that, if positive FCIs are derived by a non-implicature mechanism while NFCIs are genuine implicatures, then children should pattern with adults on positive FCIs but diverge more clearly on NFCIs, in a way that aligns with what we know about their behavior with scalar implicatures.

This is not what we find overall. In Experiment 1, children compute NFCIs at ceiling, nearly as often as positive FCIs, and in fact more often than adults do. While children’s performance on NFCIs matches their performance on negative SIs, however, a challenge of interpreting this result is that both are computed at much higher rates than those observed in adults. In the case of Experiment 1, the hybrid approach could in principle shift the burden of explanation to negative scalar implicatures, with the hope that whatever accounts for children’s high rates in that condition could also account for their performance on NFCIs.

This strategy, however, does not extend straightforwardly to Experiment 2. Here, children (as

with our adult participants) did not compute NFCIs. Crucially, however, NFCIs no longer pattern with negative scalar implicatures, which show instead a pattern more similar to that observed in Experiment 1. Taken together, these results do not yield the clear developmental alignment between negative FC and negative SIs that the hybrid approach would lead us to expect.

Taken together, then, what conclusion should be drawn from these experiments? Overall, the results of Experiment 2 introduce a new challenge for implicature-based approaches to NFC, and indirectly for the hybrid account. On the one hand, one might conclude that the hybrid response to the acquisition and polarity challenges is not correct; but in doing so, one would allow the original problems for implicature-based accounts to re-emerge in full. On the other hand, one could maintain the hybrid response; an additional explanation, however, would be needed for why NFC behaves differently across the modal and nominal domains, that is, for the contrast between the high rates of NFCI computation in Experiment 1 and the absence of comparable evidence in Experiment 2. In the first case, the difficulty is specific to the hybrid approach; in the second, it becomes a more general problem for implicature-based accounts of NFC. We therefore conclude that, although our findings replicate the polarity asymmetry motivating the hybrid account, the developmental pattern observed in children raises a substantial challenge for that account and, more broadly, for SI-based approaches to NFC.

That said, it is also not clear to us that a non-implicature account of both FCIs and NFCIs could straightforwardly accommodate our results. In principle, one might explain the asymmetry between FCIs and NFCIs in adults by positing a mechanism that is available to adults but not yet to children. Such a move could, in turn, also account for the difference between adults and children. For example, a presuppositional account of FCIs and NFCIs, such as the one advanced in Del Pinal et al. (2024), might attribute the adult asymmetry to a difference in how easily the relevant inferences are accommodated. Such an account could then argue that accommodation is generally difficult for children (see Bill et al. 2016 for discussion), thereby predicting no difference between FCIs and NFCIs in the latter group.¹⁵ A similar line of reasoning might be pursued for a cognitive-bias account along the lines of Aloni (2022). However, in order to fully account for our findings, both approaches would also have to be complemented with an account of the differences between positive and negative scalar implicatures.¹⁶

5.2 Open issues

Before concluding, we briefly note three additional aspects of our results which are not central to our main investigation, but which are nonetheless unexpected and deserve further attention.

¹⁵This option is not available to the otherwise similar account proposed by Goldstein (2019), since that account does not predict NFCIs.

¹⁶An independent proposal in the literature holds that negative SIs can arise in two ways (Ramotowska et al. 2024; Amiraz 2022): (i) as a scalar implicature of the sentence, or (ii) as an inference associated with the implicit question under discussion. More specifically, the idea behind (ii) is that speakers are biased to interpret negative sentences as addressing their corresponding polar question. For instance, an utterance of *Mary doesn't have to buy the apple* is typically understood as addressing the question of whether Mary *has* to buy the apple; this question, in turn, may prompt the inference that Mary is allowed to buy the apple in the first place, either as a presupposition or as an implicature of the question itself (see Romoli 2012 for discussion). This dual source could help explain why the inference is relatively robust in adults and why children may access it more readily than its positive counterpart. Such an account could, in principle, be combined with either of the non-implicature approaches outlined above, and potentially also with the hybrid approach. We leave a fuller exploration of this possibility for future work.

The first concerns positive scalar implicatures. Across the two experiments, children and adults exhibited very similar rates of positive scalar implicature computation. In Experiment 1, this appears to be due to adults deriving implicatures at a lower rate than is typically reported (around 20%), whereas in Experiment 2 it seems instead that children derived them at a higher rate than expected. Either way, this provides an additional reason to test the robustness of this finding in further samples of children.

The second concerns the absence of evidence for negative free choice in Experiment 2. This finding leaves open the question of whether the inference was simply not detected in our study, or whether it is in fact absent from this domain altogether. At the same time, it is worth noting that this null result arises in a design that was clearly able to detect other relevant inferences in the same domain, including disjunction-based free choice inferences and regular scalar implicatures. This suggests that the burden of proof may lie more strongly with the existence of negative free choice in this domain than with the sensitivity of the experimental design itself (see Marty et al. 2024a for relevant discussion).

Finally, in Experiment 2, both adults and children showed a relatively high rejection rate for the true control in the FC conditions. In particular, a non-negligible proportion of adults rejected this control condition in the FCI and low FC cases (around 20%), and children did so even more. While this issue merits further investigation, one possible explanation is that participants were deriving an additional inference on top of the free choice inference itself. In the FCI case, for instance, once participants derived the inference that some girls bought the apple and some girls bought the pear, they might also have inferred that, in each case, the group involved was plural—that is, that more than one girl bought the apple and more than one girl bought the pear. This further inference, sometimes called a *multiplicity* inference in the literature, was in fact false in our control condition, both for FCI and for low FC. We leave a fuller investigation of this possibility for future work.

6 Conclusion

This paper examined negative free choice inferences from both an experimental and a developmental perspective. Our results replicate the asymmetry between positive and negative FCIs previously observed in adults, and extend the finding to the nominal domain. The results also show that children are broadly adult-like in displaying weaker rates of negative FCIs compared to positive FCIs. At the same time, the developmental pattern does not fit neatly with the predictions of the hybrid approach, since children do not show the expected stable alignment between negative FCIs and negative SIs. We therefore conclude that, while the hybrid account captures an important polarity asymmetry, it does not yet provide a satisfactory explanation of the developmental facts. More broadly, our findings highlight the need for a theory of free choice that can accommodate both the adult polarity contrast and the acquisition data.

A Instructions for Experiment 1

We're going to play a game with our friend Fluffy.



In each story, Fluffy will tell us something about one of two children, Sam and Mia. Here they are:



Sam



Mia

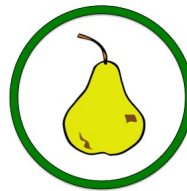
[Click to continue]

Sam and Mia are shopping at the store.

Their parents have some rules about what Sam and Mia are allowed to buy, what they have to buy, and what they are not allowed to buy. Here is how we'll represent these rules:



The **red circle** with the line through the burger means that they are not allowed to buy the burger.



The **green circle** around the pear means that they are allowed to buy the pear, but they don't have to buy it.



The **black square** around the banana means that they have to buy the banana.

[Click to continue]

You'll see some pictures depicting these rules. Each time Fluffy will say what Sam or Mia can or cannot buy according to these rules. Your job is to decide if Fluffy is right or wrong.

(Adult participants: you will click on 'Yes' for right, and 'No' for wrong)

Let's try some examples together!

_____ [Click to continue] _____

Let's meet Fluffy now! Fluffy, can you see us?

[video clip of the puppet saying: 'Hi there! I can see you!']



Are you ready to play the game, Fluffy?

_____ [Click to continue] _____

[video clip of the puppet saying: 'Yes! Let's go!']



Let's go!

B Instructions for Experiment 2

We're going to play a game with our friend Moomoo.



In this game, you'll see some pictures about boys and girls and the things that they have. Moomoo will tell us something about each picture. Your job is to decide if Moomoo is right or wrong.

(Adult participants: you will click on 'Yes' for right, and 'No' for wrong)

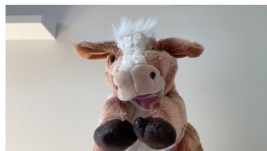
Let's try some examples together!

Example practice trial

Look at this picture:



Let's say Moomoo says the following:



Is Moomoo right?

Yes

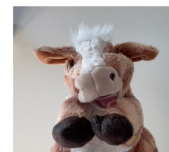
No

(a) Item presentation

Look at this picture:



Let's say Moomoo says the following:



**In this case, we would say that Moomoo was right!
The girl DOES have a butterfly!**

Next

(b) Feedback

After the practice trials

Let's meet Moomoo now! Moomoo, can you see us?

[video clip of the puppet saying: 'Hi there! Yes, I can see you!']



Are you ready to play the game, Moomoo?

[Click to continue]

[video clip of the puppet saying: ‘Yes, let’s go!’]



Let’s go!

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